

CIS 613: Model-Based Testing

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Test-Driven Development vs Model-Driven Development

Model-Driven Development is a rigorous, top-down approach.



Test-Driven Development is a bottom-up approach.

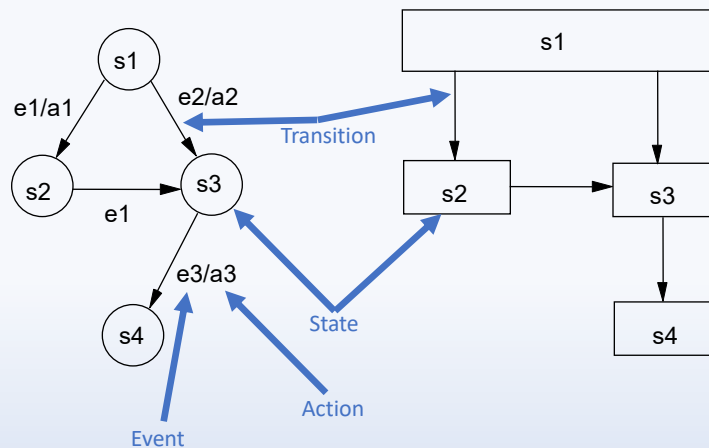
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One Method of Modeling Behavior: Finite State Machine (FSM)

- Components:
 - states (data)
 - transitions (actions)
- No Memory
- State Independence
- Represented by diagrams (graphs) or tables

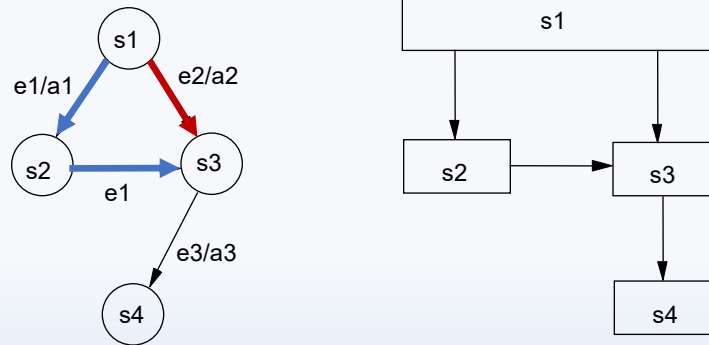
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FSM: Graph Representation



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FSM: Paths from s1 to s3?



Events are *state* dependent!

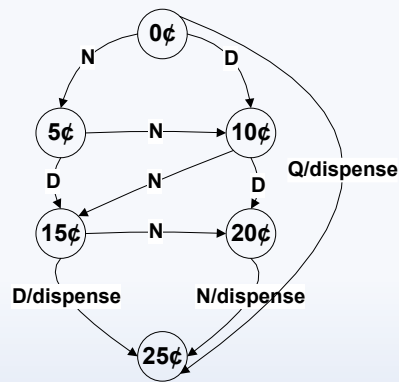
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FSM: What is a State?

A state is an **interval of time** in which some state **proposition** is true.

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FSM: Antique Vending Machine



Event sequences:

1. N D D
2. D N D
3. D D N
4. N N N D
5. N N D N
6. N D N N
7. D N N N
8. N N N N N
9. Q

What happens if we put in three
dimes? (D D D)

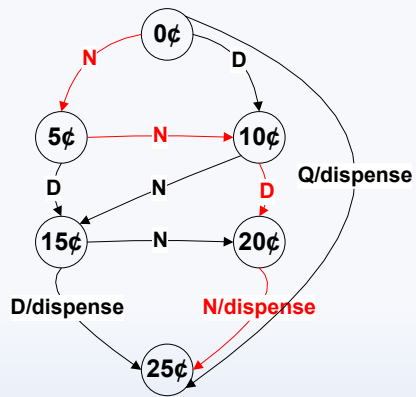
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State Propositions

- What are the state propositions for the Antique Vending Machine FSM?
- What relationship, if any, exists among these state propositions?
- How does a true state proposition become false?
- What is the state proposition of a program flow diagram?

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Testing a FSM



Event

sequences:

1. N D D
2. D N D
3. D D N
4. N N N D
5. **N N D N**
6. N D N N
7. D N N N
8. N N N N N
9. Q

Test Procedure:

1. Deposit nickel (**N**)
2. Deposit nickel (**N**)
3. Deposit dime (**D**)
4. Deposit nickel (**N**)
5. **Verify** item dispenses

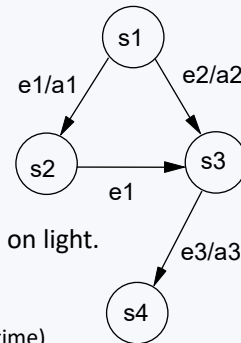
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FSM: Execution

- Stream of **input events** causes a **series of transitions**
- States are **reachable** and **unreachable** based on **input events**
- Legitimate and impossible event sequences
- Software Safety (never permit a system to remain in a hazardous state)

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In Class Exercise!



Make a Finite State Machine for motion based lights in a classroom. **Note:** if switch is off, motion won't turn on light.

Input events: (M, S_ON, S_OFF)

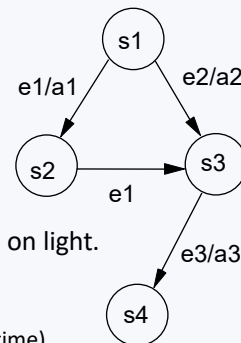
- M = Motion Detected / Not Detected (for some period of time)
- S_ON = Switch Turned On
- S_OFF = Switch Turned Off

Output events (ON, OFF):

- Lights Off
- Lights On

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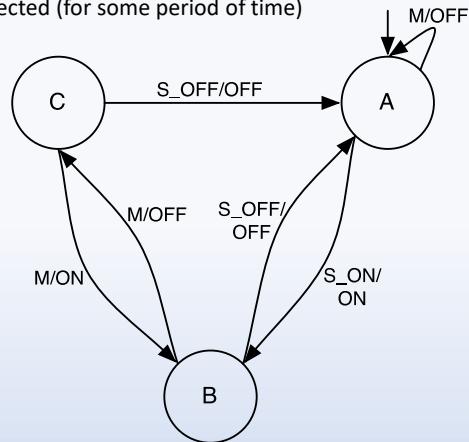
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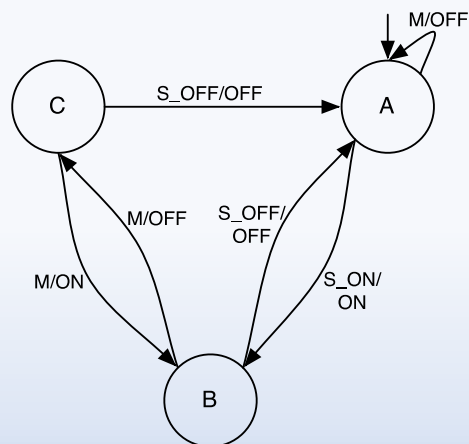
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In Class Exercise!

- Create a set of test cases to cover:

- Edge Coverage, and
- Node Coverage

- What is the difference?

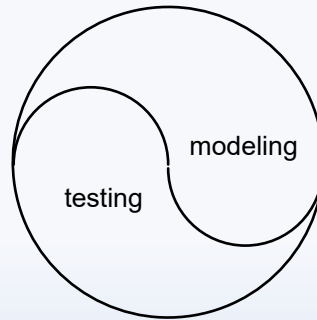


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Overview of Model Based Testing (MBT)

An iterative process:

1. Start by modeling the System
2. Derive test cases from the model
3. Execute test cases
4. Record coverage
5. Trace to model
6. Modify model as needed
7. Repeat



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Modeling the System

- Models were developed to meet the needs of the systems being analyzed
 - Issues inherent in the system must be expressed
 - In the book, Petersen's Lattice presents a nice selection (but dated) of behavioral models
- The hard part is picking the *right* model,
 - Not too powerful (hard to analyze)
 - Not too weak (hard to express)

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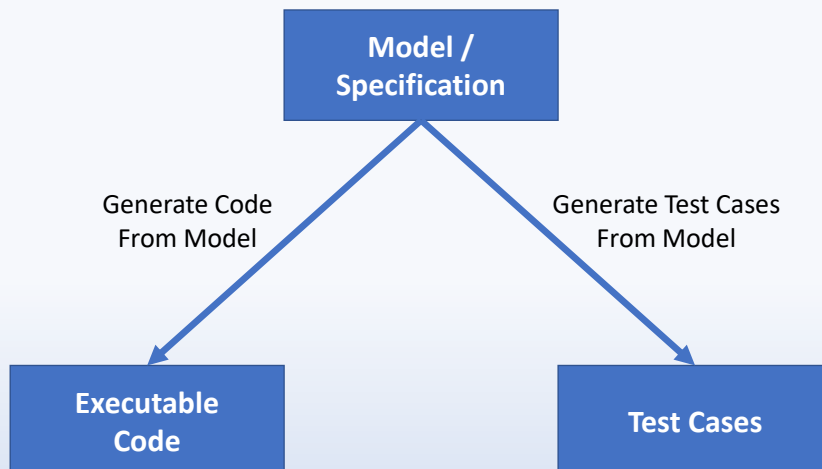
“Procedure” for Model-Based Testing

- Understand application to be modeled
- Identify behavioral issues in the application
- Select *appropriate* model
 - this may depend on staff talent
 - or existing development tools
 - or other immutable requirements
- Model the application
- Use an engine to “execute” the model, thereby identifying
 - interesting scenarios
 - automatically generated test case (skeletons)

Lots of tool support - Could be good topic to look for a project!

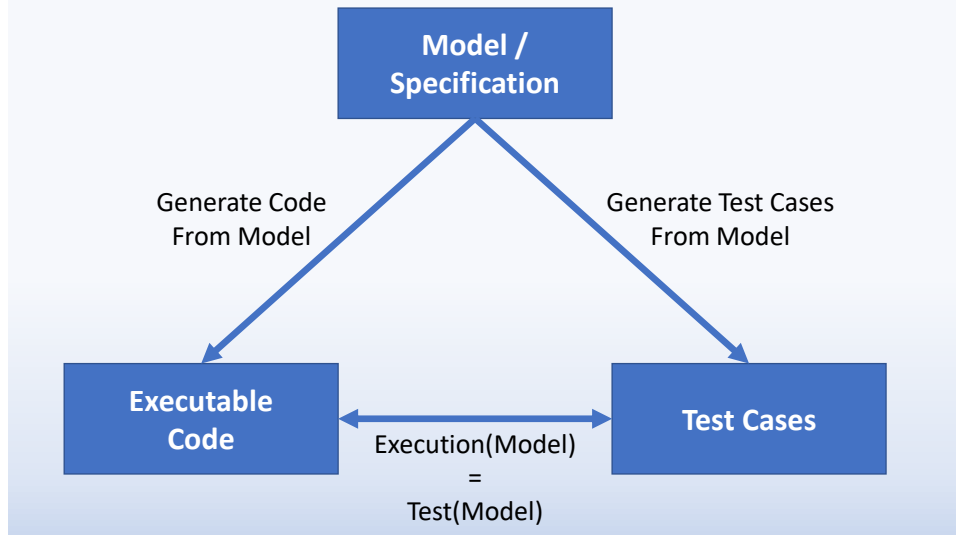
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Executable Specification vs Model-Based Development vs Model-Based Testing?



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